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Graph-based networks for accurate prediction of ground and excited state molecular properties from minimal features†

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Graph neural networks (GNN) have been demonstrated to correlate molecular structure with properties, enabling rapid evaluation of molecules for a given application. Molecular properties, including ground and excited states, are crucial to analyzing molecular behavior. However, while attention-based mechanisms and pooling methods have been optimized to accurately predict specific properties, no versatile models can predict diverse molecular properties. Here, we present graph neural networks that predict a wide range of properties with high accuracy. Model performance is high regardless of dataset size and origin. Further, we demonstrate an implementation of hierarchical pooling enabling high-accuracy prediction of excited state properties by effectively weighing aspects of features that correlate better with target properties. We show that graph attention networks consistently outperform convolution networks and linear regression, particularly for small dataset sizes. The graph attention model is more accurate than previous message-passing neural networks developed for the prediction of diverse molecular properties. Hence, the model is an efficient tool for screening and designing molecules for applications that require tuning multiple molecular properties.

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We present general-purpose graph neural network-based models for accurate prediction of a wide range of molecular properties, including orbital energy and optical properties, and we show them to be invariant with respect to dataset size and origin. The models resolve molecular structures into graphs and effectively learn correlations between structural features and properties. We show that adding an attention-based mechanism in the learning process consistently leads to the highest accuracy, and outperforms regression-based learning. Optimal pooling methods are implemented in the convolution layer of the network to achieve high prediction accuracy for both ground state and excited state properties. The model enables the design of organic molecules with a required set of electronic and optical properties by predicting diverse properties with high accuracy based on minimal information regarding molecular structure as input.

1 Introduction

Accurate prediction of molecular properties by computational methods can inform decision-making and accelerate the discovery of novel materials and compounds with tailored properties. Prediction of molecular properties with computational methods and correlations of these properties with molecular structure enables rapid assessment and

screening of molecules having required properties.^{1–3} Several characteristics of molecules involving energy levels, orbitals, and excitation transitions are known to influence their behavior and, consequently, performance in a given application. While molecular features such as size, elements, and bond types are known to determine molecular properties, their effects combine in complex ways to constitute molecular behavior. Further, different properties are influenced more heavily by certain features than others. As such, a direct correlation between combinations of geometric factors and various properties has yet to be established, and predicting a range of molecular properties based on minimal information regarding molecular structure remains a challenge.

Prediction of ground state as well as excited state molecular properties to a high degree of accuracy has been made possible by first-principles methods.^{4–7} These methods

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involve an approximate solution to Schrödinger's equation for the molecule, thereby obtaining energy values for different molecular states. Approximations are in the form of the Hartree-Fock method, which solves for individual orbitals to combine them for a composite solution for the molecule,⁸ or by methods employing density functional theory (DFT) whereby the electronic density is optimized as a means of rapidly obtaining the corresponding ground state of the molecule.⁹ However, first-principle calculations are time-consuming and computationally demanding.

Graph neural networks (GNN) have emerged as a novel solution recently.^{10–16} Features of molecular geometries, including information regarding atoms and bonds, are encoded as graphs consisting of node and edge embeddings, on which networks are trained to generate linear functions thereby correlating structural features of molecules with their properties.^{17–22} Graph convolution networks (GCN) are a case of GNN where convolution layers of the network process the input by performing message passing before the network learns its correlations with the target values. Additionally, the convolution layers in graph-based models pool the updated graph embeddings to generate a final molecular embedding vector. The pooling method provides an additional avenue to optimize the weighing of input features. Previous attempts have been made to correlate orbital energies, states, and optical properties with molecular structures using graph-based and machine-learning methods.^{23–27} However, these have relied on numerous structural features, making them computationally expensive. Further, no versatile algorithms simultaneously predict the orbital energy and optical absorption or emission wavelengths of molecules, as there are differences in the influence of molecular features on the ground state and excited state behavior.^{28–33} Due to this, graph neural networks that predict both ground state and excited state properties have largely not been developed.^{19,34}

Recently, graph attention (GAT) networks have been developed to utilize attention mechanisms to aggregate information from neighboring nodes during message passing selectively. Here, node embeddings are modified based on their neighbors' attention coefficient values and updated based on subsequent neighbors' values. This exchange of information between direct and indirect neighbors captures local and global patterns of atomic interactions.³⁵ In this way, GAT models have the advantage of processing the input by weighing the contributions of the various nodes in a molecular graph. While GAT models can distinguish between features in the form of weights suitable for different targets, versatile GAT networks that are accurate for general targets have not been explored. For example, Buterez *et al.* and Na *et al.* have developed attention mechanisms tuned to predict molecular orbital energies with high accuracy.^{14,36} However, GAT models that predict optical properties alongside energy within a fixed architecture have not been developed.

In this work, we present an optimized graph-based neural network for enhanced molecular property prediction for ground state as well as excited state properties. Molecular

properties such as energy levels of frontier orbitals, *i.e.*, highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO), along with ionization potential (IP), determine electrical conductivity and reactivity of the materials. We show that the GAT model predicts these properties with high accuracy. In addition, excited-state properties such as wavelengths of maximum absorption (λ_{abs}) and emission (λ_{em}) sourced from a different dataset and vertical excitation energy (VEE) determine the optical and chemical properties of the material. While the global pooling method is generally implemented in the convolution layer, we also explored the effect of hierarchical pooling on the prediction of properties due to its additional scoring system.³⁷ Hierarchical pooling was implemented in the graph processing to optimize the model for predicting excited state properties with as high an accuracy as observed for ground state ones. While hierarchical pooling has been used widely to tune machine learning models, its effectiveness in predicting molecular properties has not been demonstrated. The graph-based models were compared to a linear regression method directly correlating compressed input embedding to target properties based on a random forest algorithm. Our results reveal the consistent performance of the graph-based models over the random forest approach in predicting these diverse experimentally and computationally determined molecular properties for organic molecules, with the GAT approach leading to maximum accuracy. The graph-based models presented here are accurate and applicable for diverse properties and open avenues for further advancements in molecular informatics.

2 Computational details

Fig. 1 shows the workflow to run the model for property prediction. The dataset comprises molecular structures in SMILES strings and corresponding values of measured or computed properties. Relevant information regarding the molecular structure is encoded as one-hot embeddings. In graph-based methods, a convolution network updates these to yield a pooled embedding ($\vec{G}$) for each molecule, which is then subsequently fed to a feed-forward network layer for learning correlations with target property. A loss between the target property and the predicted value is then backpropagated to obtain updated weight matrices for the entire end-to-end model.

2.1 Dataset

We aimed to validate the developed model for several molecular properties relevant to various applications. Six molecular properties were the focus of our study, HOMO, LUMO, IP, VEE, λ_{abs} , and λ_{em} . Two separate databases were the data sources for different properties (set-I & set-II). Set-I contains the ground state and excited state properties—HOMO, LUMO, IP, and VEE, as computed *via* first principles for 1709 organic molecules, while set-II consisted of experimentally measured λ_{abs} and λ_{em} of 6085 distinct

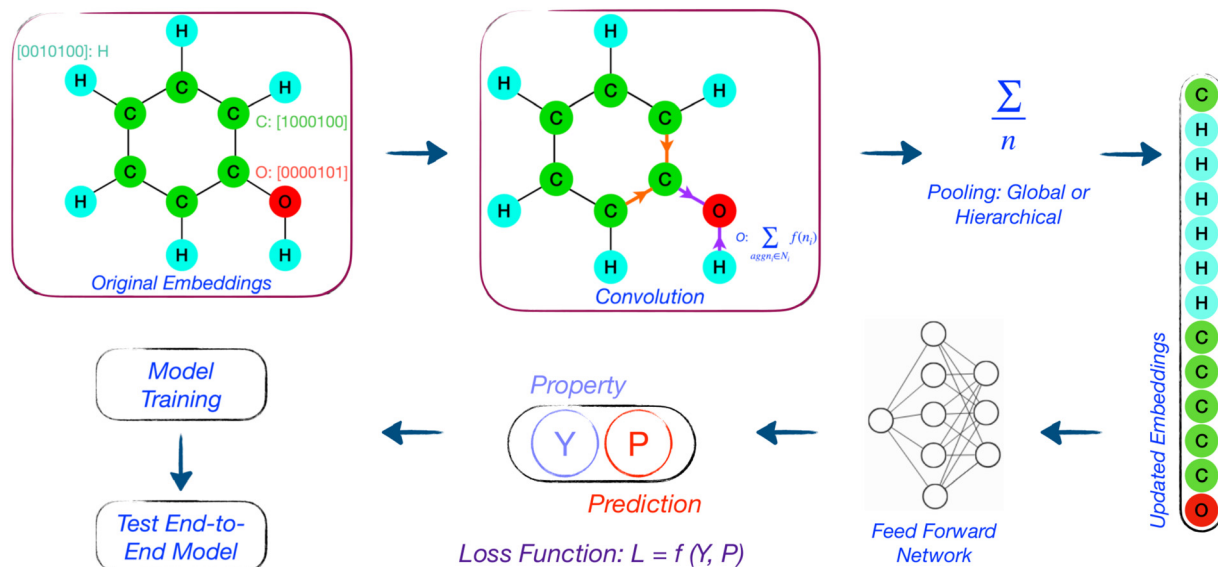


Fig. 1 Scheme displaying workflow of the procedure. Molecular structures are encoded using one-hot embeddings utilizing information regarding atoms and bonds. In graph-based methods, these are then updated by a convolution network and passed to a linear layer for a downstream prediction task. The predicted value is compared with the experimentally measured (or computed) value based on a predefined loss function, which is minimized by backpropagation. Hyperparameters are optimized to drive the process efficiently before being validated with test data.

organic molecules. The difference in size allows us to compare the performance of models with a smaller and larger dataset. Set-I was obtained from Virtual Excited State Reference for the Discovery of Electronic Materials Database (VERDE Materials DB) accessible at URL “<https://www.verdematerialsdb.com/>”³⁸ computed by DFT-based optimization and single-point calculations. Set-II was obtained from a Nature Scientific database as a compilation of experimentally measured optical properties, λ_{abs} and λ_{em} .³⁹ Set-I and set-II are independent and have vastly different sizes, thus allowing us to verify the robustness of the model under variations in dataset size. Both databases consist of π -conjugated organic molecules with notable optical properties and have applications ranging from solar cells and light-emitting diodes to redox flow batteries. Fig. 2 shows the size distribution amongst the molecules in either of the

databases according to number of atoms constituting each molecule. The size of molecules ranges from less than ten atoms to 75 in set-I and 150 in set-II. Both sets have an extensive range of molecular sizes, while the distribution varies, and are sufficiently randomized. However, in set-II, while the range is extended to molecules with up to 150 atoms, the number of large-sized molecules is low. Therefore, some structural features of large molecules have less occurrence, leading to a skew in molecular information. Fig. 3 shows the corresponding distribution in molecular property values. The skewness of the dataset was calculated for all cases using the statistics module of SciPy library in Python language. It is found that the skewness is highest for data of λ_{em} (1.17), followed by that of HOMO (−0.95). Skewness is moderate for IP (0.64) and λ_{abs} (0.84), while it is low for LUMO (−0.24) and VEE (−0.24). Variations in the features within the dataset are shown in Fig. S1 of the ESI.† Both datasets have a similar distribution of features, *i.e.*, molecules with fewer unsaturated bonds than saturated bonds and more atoms in the form of sp^2 hybridization. Further, in both datasets, the distribution of aromaticity is closely analogous to molecular size distribution, as shown in Fig. 2. This indicates that the molecular structures are composed mainly of aromatic rings, as is the case for conjugated organic molecules. We note that the feature distribution, and hence the input to the model, are similar in both datasets, implying that the main distinction between them is dataset size. Accordingly, a difference in the performance of the model with respect to the dataset has been attributed to a response of the model to a difference in the size of the input rather than to their features.

Initially, structural information of the molecules was extracted from their SMILES strings and converted in the

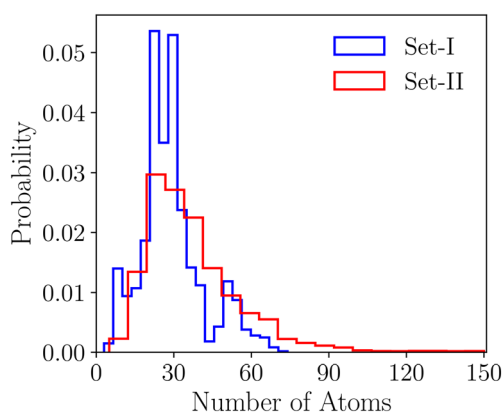


Fig. 2 Distribution of sizes of molecules in both datasets displaying a larger skew in set-II.

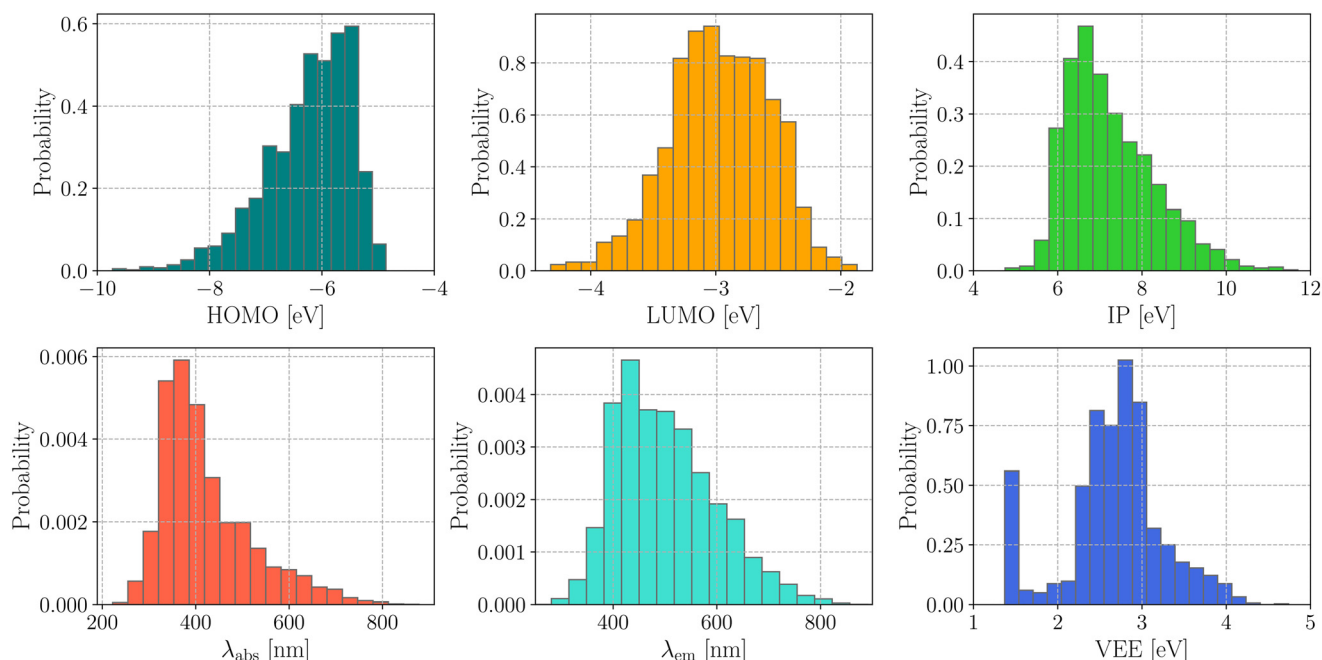


Fig. 3 Distribution in the absolute values of various properties trained (tested) in this study.

form of a graph representation.⁴⁰ As shown in Fig. 1, the SMILES strings are at first represented in the form of one-hot embeddings (original embeddings) containing information about atoms in the form of nodes and bonds in the form of edges. The one-hot embeddings are vectors with elements with binary values that are set to zero by default and switched to one at those indices within the vector that is assigned to the particular feature present. This information is mainly encoded in two parts: atomic information, such as atomic number, is contained in node embedding matrices, along with information including charge, aromaticity, and orbital hybridization. Edge embeddings contain information on bonds, including bond type, the stereochemistry of the bond, and its presence in a cyclic structure. The attributes and length of vectors corresponding to each of the respective embeddings are given in Table 1. The original embeddings were generated by RDKit package,⁴¹ run in Python.

Table 1 Details of node and edge embeddings implemented in the graph-based model

Node attributes	Node embeddings
Atomic number	118
Atomic charge	5
Atomic degree	6
Hybridization	5
Aromaticity	1
Edge attributes	Edge embeddings
Bond type	5
Bond stereo	6
Bond rings	1

2.2 Learning models

2.2.1 Random forest. Random forest model has been employed here as the linear regression method. Random forest is an ensemble learning method⁴² that combines the output of multiple decision trees to predict a target value. The one-hot embeddings obtained from SMILES strings were first converted to an average embedding vector for each molecule. Target values for the respective properties were processed by RobustScaler⁴³ of the scikit-learn package to make them suitable for regression. The embedding and target values were passed as inputs to a random forest regressor model.

2.2.2 Graph convolution network. The graph convolution networks method processes the original embeddings of molecules to obtain an updated graph more correlated to the target properties. Original embeddings extracted from SMILES strings were input to two convolution layers, where the values of nodes (h_i) are updated according to those values of nodes [$h_j, j \in \mathcal{N}(i)$], where $\mathcal{N}(i)$ is the neighborhood of node i . Updating of the nodes is carried out as per eqn (1).

$$\vec{h}_i^{\text{new}} = \vec{h}_i^{\text{old}} + \text{ReLU} \left(\sum_{j \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(i)| \cdot |\mathcal{N}(j)|}} \cdot W \cdot \vec{h}_j^{\text{old}} + b \right) \quad (1)$$

Here, W is the weight matrix, and b is a bias. Matrices W and b were updated in every epoch to attain minimum loss function and thus are central to the learning process. The non-linear activation function known as a rectified linear unit (ReLU) was applied to obtain the final updated value of the node h_i . In the second convolution layer, the values of h_j that are now input for every h_i as per eqn (1) have been updated

by their neighbors, the second-order neighbors for h_i . The updated embeddings h_i^{new} were converted to the molecular embedding vector $\vec{G}$ by pooling operations.

2.2.3 Graph attention network. GAT is a graph-based method³⁵ where attention mechanisms are used to weigh neighboring nodes' contributions during message passing to learn the strength of correlations. This was done by weighing the attention scores of all h_j to determine their influence in updating the value of h_i . The scores result in attention coefficients (α_{ij}) for connected nodes (i, j), which were applied to update h_i . The coefficients were calculated as per eqn (2).

$$\alpha_{ij} = \frac{\exp\left(\text{ReLU}\left(W \cdot \vec{h}_i \| \vec{h}_j\right)\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\text{ReLU}\left(W \cdot \vec{h}_i \| \vec{h}_k\right)\right)} \quad (2)$$

Here, h_i^{new} was then obtained as a weighted sum of the features of h_j , based on α_{ij} as per eqn (3).

$$\vec{h}_i^{\text{new}} = \text{ReLU}\left(h_i^{\text{old}} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \vec{h}_j^{\text{old}}\right) \quad (3)$$

Hence, the GAT model has an additional mechanism of learning α_{ij} for all (i, j), which tunes the message passing between different neighboring nodes. The implementation of the model is similar to GCN, in a batch process with the original embeddings updated by the message passing layers and pooled to form $\vec{G}$.

2.3 Implementation

The models are trained for each property as per the methods described above. Two pooling methods were compared for generating $\vec{G}$ by the graph-based methods: a global mean pooling and a hierarchical pooling. The global mean pooling consists of two convolution layers, the readout of which is concatenated and compressed to generate $\vec{G}$. In the case of hierarchical pooling, after a convolution layer has performed message passing, a score is generated for the processed embedding so that the nodes whose value exceeds the score are passed onto the next convolution layer.³⁷ This process is repeated twice, and the readout from all layers is summed to generate the molecular embedding $\vec{G}$. $\vec{G}$ is passed as input to the subsequent linear layer for learning. Model parameters such as weight matrices and activation functions were learned during training. Both graph-based models were run as an end-to-end batch process, updating the convolution and linear layer parameters with each epoch. The batch size, learning rate, and bias hyperparameters were optimized for each target property, *i.e.*, each molecular property. The graph-based networks were implemented based on functions available with PyTorch.⁴⁴

All three algorithms seek to minimize the loss function, which is a measure of the difference between predicted ($\hat{y}_i$) and actual values (y_i). The loss function adopted for all

cases was the predefined smooth L1 loss. A transition function is adopted to attain the advantage of mean and mean squared error modes of representing loss functions, as given in eqn (4).

$$\text{SmoothL1}(\hat{y}_i, y_i) = \begin{cases} 0.5(\hat{y}_i - y_i)^2 & \text{if } |\hat{y}_i - y_i| < 1 \\ |\hat{y}_i - y_i| - 0.5 & \text{otherwise} \end{cases} \quad (4)$$

The SmoothL1 Loss function was applied element-wise to calculate the loss for each element in the batch, whereby a mean was applied to obtain the final loss value of the batch.

For each of the properties, the model was trained with 80% of the data, partitioned after performing a random shuffle of the dataset. Adam optimizer was applied, and batch normalization was utilized to prevent overfitting. Hyperparameters, including batch size and learning rate, were fine-tuned based on their impact on validation performance. To comprehensively evaluate our model's predictive performance, we employed standard evaluation metrics: mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and linear correlation coefficient (r^2). As MAPE and r^2 are relative quantities, they are compared across properties and models. These metrics encompass diverse dimensions of predictive accuracy, ensuring a thorough assessment of our model's effectiveness.

3 Results

3.1 Ground state properties

The ground-state molecular properties studied here include HOMO, LUMO, and IP. The target values were obtained from set-I for 1709 molecules. Fig. 4 shows the prediction capabilities of the GAT model for properties HOMO, LUMO, and IP. IP was calculated as the difference between molecular energy in the neutral ground state and the positively charged state. As it does not depend on the molecule's excited state, it is classified with ground state properties. The linear functions obtained by training the model for each property are displayed, along with the predicted values plotted against actual values as a scatter. The critical evaluation metrics MAE, RMSE, MAPE, and r^2 for all methods implemented are tabulated in Table 2. MAPE and r^2 are compared across properties as they are relative quantities. It can be seen that the graph-based methods achieve high accuracy for all properties. In the case of the GAT model, values of r^2 for the prediction of unknown values during testing are found to be 0.95, 0.93, and 0.86, respectively, for IP, HOMO, and LUMO, asserting the efficiency in predicting values of orbital energy based on only molecular structure information. MAPE values for IP, HOMO, and LUMO are 2.44%, 2.52%, and 3.79%, respectively, further reinforcing the model's efficiency. The model displays high accuracy in predicting ground state properties despite the small size of the dataset.

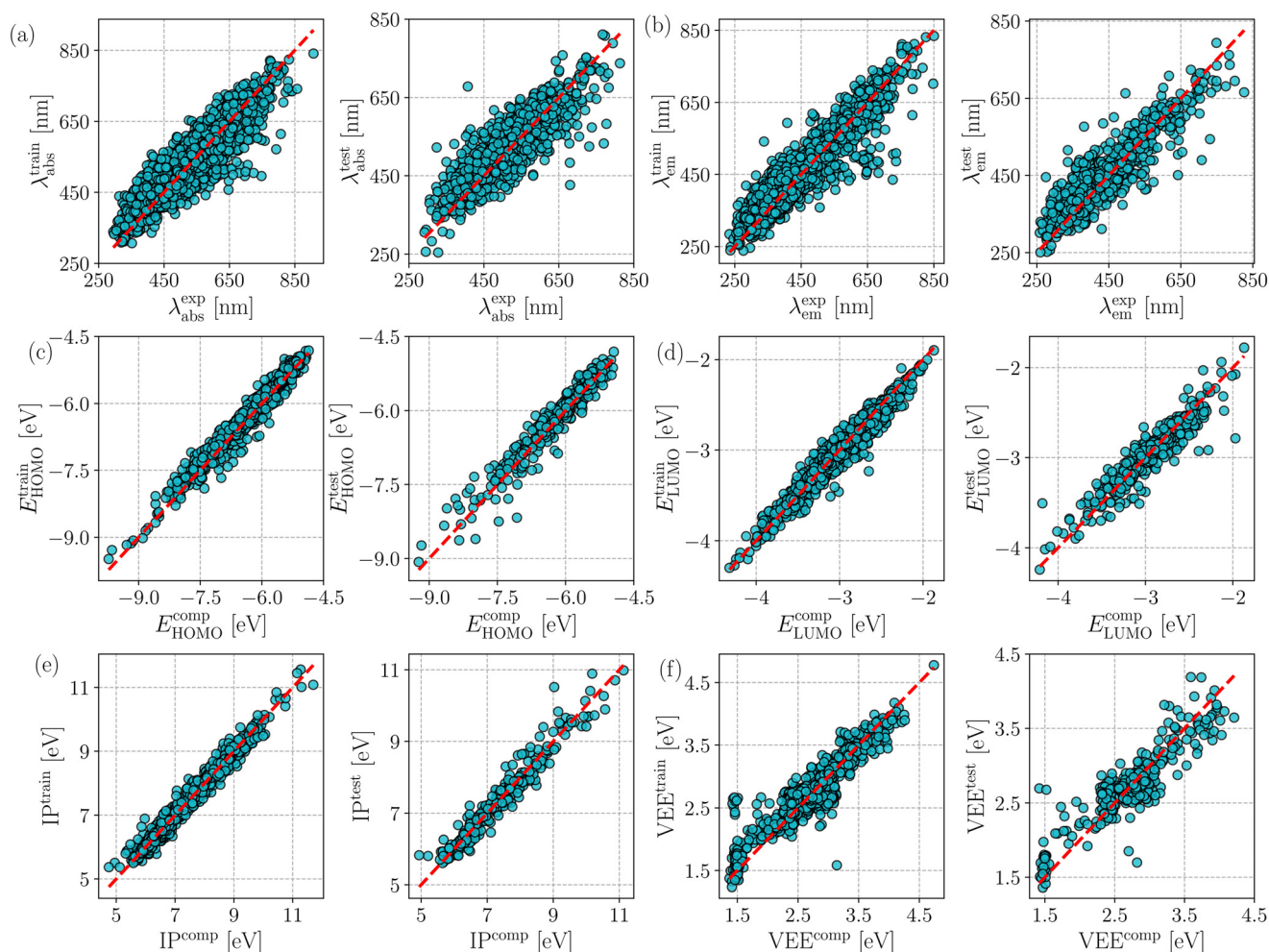


Fig. 4 Predicted properties *versus* their actual values, along with trained function for training and testing steps for diverse molecular properties: (a) λ_{abs} , (b) λ_{em} , (c) HOMO, (d) LUMO, (e) IP, and (f) VEE.

Table 2 displays the performance metrics for predicting ground state properties by graph-based and RF models. Values of all metrics are slightly better for the GAT model than GCN. IP is predicted with high accuracy, r^2 is 0.95 and MAPE 2.44% with GAT, while r^2 and MAPE are 0.92 and 2.73% with GCN. The difference in performance is more significant with random forest, where the r^2 and MAPE for prediction of IP are 0.86 and 3.81%. The GAT model consistently predicts values with marginally higher accuracy than GCN, and both graph-based methods perform better than random forest. While the accuracy of RF is known to be sensitive to data size, we show graph-based methods to be size-adaptive and highly accurate even for smaller data sets. Both graph-based algorithms process the original embeddings to generate G for learning. The updating process with GAT involves learning coefficients weighing connections amongst nodes as given in eqn (2). However, the differences in prediction accuracy among both graph methods are minor, indicating that the specific atoms and their neighbors are a more significant source of information for the network than their relative connections. Graph methods outperform

RF even when the dataset size is small, as is the case for set-I. Additionally, to verify the general applicability of the GAT model for molecular design, we applied it to a separate dataset of donor-acceptor molecules. As detailed in Section S1 of the ESI†, the performance metrics from these tests further confirm the model's effectiveness in accurately predicting key properties, demonstrating its potential as a reliable tool for designing new molecules.

3.2 Excited state properties

The graph-based models developed were further trained to predict excited state properties. The excited state properties include VEE, values of which are obtained from dataset set-I. Further, λ_{abs} and λ_{em} were studied, and the target values were obtained from experimental values in set-II for 6085 molecules. In this way, diversity is imparted in predicting properties by optimizing for different datasets having variations in data size and sources. As shown in Table 2, r^2 and MAPE for VEE are 0.80 and 7.72%. The values of r^2 for properties from set-II, λ_{abs} and λ_{em} , are 0.71 and 0.78,

Table 2 Performance parameters, RMSE, MAE [in nm unit for λ_{abs} and λ_{em} , whereas in eV unit for the remaining properties], MAPE [%], and r^2 , for prediction of respective molecular properties for algorithms based on RF, GCN, and GAT

Property	Model	Metrics			
		RMSE	MAE	MAPE	r^2
E_{HOMO}	Random forest	0.32	0.22	3.33	0.83
	GCN	0.27	0.2	3.16	0.88
	GAT	0.22	0.16	2.52	0.93
E_{LUMO}	Random forest	0.21	0.14	4.62	0.73
	GCN	0.15	0.11	3.8	0.85
	GAT	0.15	0.11	3.79	0.86
IP	Random forest	0.39	0.28	3.81	0.86
	GCN	0.29	0.19	2.73	0.92
	GAT	0.26	0.18	2.44	0.95
VEE	Random forest	0.33	0.23	9.34	0.66
	GCN	0.3	0.23	9.48	0.74
	GAT	0.26	0.19	7.72	0.80
λ_{abs}	Random forest	52.23	38.38	7.79	0.70
	GCN	57.55	43.18	8.75	0.65
	GAT	52.19	39.68	8.10	0.71
λ_{em}	Random forest	44.51	30.13	7.18	0.79
	GCN	46.78	32.49	7.69	0.77
	GAT	45.19	33.44	8.26	0.78

respectively, and MAPE values are 8.10% and 8.26%, respectively. While the graph-based models are well optimized for ground state properties, the prediction accuracy of the excitation properties is found to be lower than that of HOMO, LUMO, and IP. This reduction in prediction quality is despite the larger dataset size of set-II corresponding to λ_{abs} and λ_{em} . As excited state properties involve electronic transitions, factors influencing these properties can differ from ground-state ones. Updating embeddings with a mechanism that prefers specific nodes can be expected to improve model accuracy. For this purpose, hierarchical pooling was implemented in the convolution layers. Hierarchical pooling enables the weighing of nodes in embedding during message passing to generate $\vec{G}$ instead of the conventional global pooling, which relies on the direct compression of updated embeddings to $\vec{G}$. In this way, the nodes that correlate more strongly with the properties get identified and contribute more to the updated embedding. Table 3 shows the performance metrics for predicting excited state properties by graph-based methods with hierarchical pooling. For the GAT method, VEE, λ_{abs} , and λ_{em} are predicted with values of r^2 at 0.80, 0.71, and 0.78 with global

Table 3 Metrics for prediction of excited state properties by graph-based methods with hierarchical pooling in convolution layer to generate $\vec{G}$

Property	Model	Hierarchical pooling			
		MAPE	r^2	MAE	RMSE
λ_{abs}	GCN	8.42	0.73	31.71	43.62
	GAT	6.93	0.77	34.89	49.05
λ_{em}	GCN	7.61	0.84	41.44	53.61
	GAT	7.33	0.86	30.16	41.88
VEE	GCN	7.53	0.82	0.19	0.26
	GAT	5.91	0.87	0.14	0.21

pooling, while it is 0.87, 0.77, and 0.86 with the hierarchical pooling method. Similarly, MAPE for VEE, λ_{abs} , and λ_{em} is 5.91%, 6.93%, and 7.33% with hierarchical pooling while it is 7.72%, 8.26%, and 8.10%, with global pooling. With hierarchical pooling, the prediction accuracy for excited state properties, regardless of datasets, well exceeds that of RF. While the prediction accuracy for ground state properties had already reached high values with global pooling, hierarchical pooling is required to maximize the model performance only for excited state properties.

To further demonstrate the dataset invariance of the model, we implement the GAT model for absorption wavelength (λ_{max}) values in set-I, as given in Section S1 of ESI.† The property of absorption is common to both datasets. As values of λ_{max} are obtained from set-I, these values are computed and available in a smaller size. The value of r^2 for prediction of λ_{max} by the GAT-based model is 0.79, and MAPE is found to be 8.06%. These are highly similar to metrics of prediction of λ_{abs} by the same model. We find the GAT-based model to achieve the same level of accuracy for the given property regardless of the size and origin of the dataset. This asserts that the model is unaffected by the size and origin of the dataset.

4 Discussion

Both graph methods outperform the RF model in terms of accuracy and precision. Convolution layers in GCN and GAT process the embeddings to generate $\vec{G}$, which correlates better with the target properties, thereby enhancing the learning process. The effect of updating original embeddings by the graph-based model is illustrated *via* *t*SNE (*t*-distributed stochastic neighbor embedding) plots in Fig. 5 for excited state property λ_{em} and ground state property HOMO. The *x* and *y* coordinates of the plots correspond to configuration coordinates obtained from *t*SNE by compressing high-dimension one-hot embeddings of the molecules to two dimensions. A color scheme is used to illustrate the values of HOMO (set-I) and λ_{em} (set-II), respectively. Fig. 5a is based on embeddings updated by the GAT model trained for λ_{em} , and Fig. 5b is based on embeddings generated by GCN of λ_{em} . Both models generate a gradually correlated configuration and property, where higher values of *x* coordinate generally correspond to a lower value of λ_{em} as per Fig. 5a and b. Similarly, Fig. 5c and d shows the *t*SNE plots for HOMO. While Fig. 5c is based on the data processed by GAT, Fig. 5d is based on embeddings updated by the GCN model trained for HOMO. In the case of the updating by GAT, embeddings form clusters with similar values of HOMO. On the other hand, the embeddings generated by the GCN form clusters of points with mixed HOMO values that are organized in a graded manner within clusters. These indicate the capacity of graphical models to update embeddings in a way that correlates with the target property. The correlation is enhanced regardless of the nature of the molecular property or the dataset. This also shows the

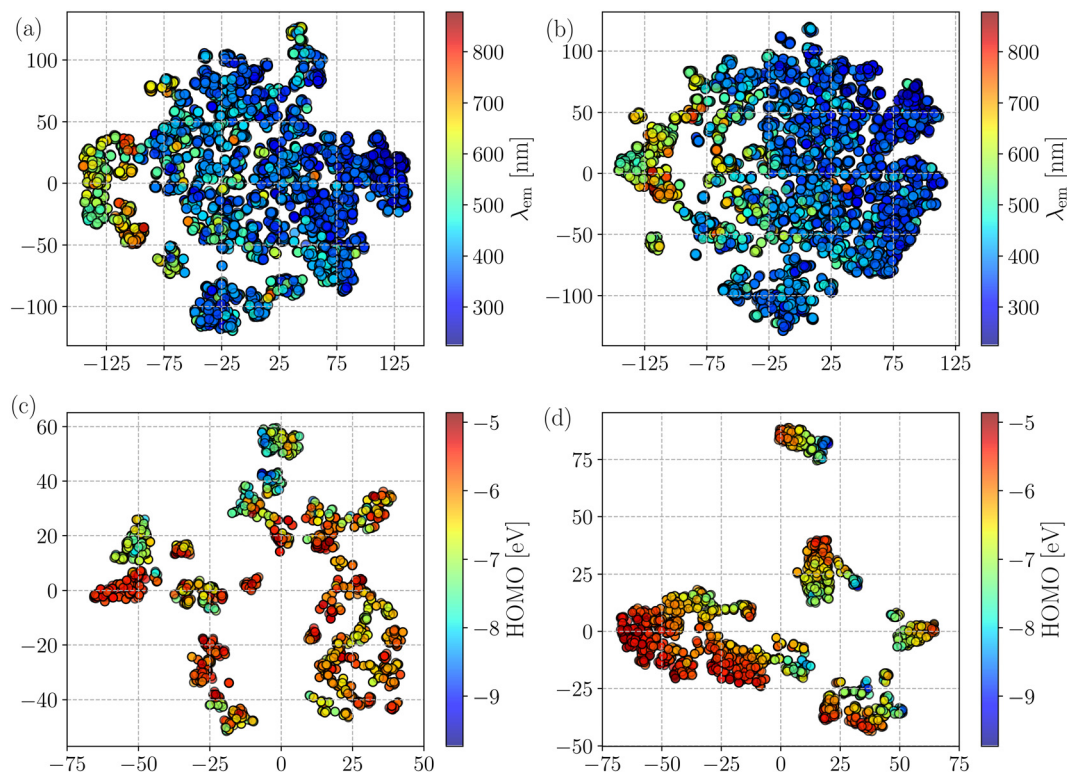


Fig. 5 tSNE plots displaying the optimal processing of embeddings for prediction of (a) and (b) λ_{em} and (c) and (d) HOMO via both graph-based approaches – GAT [(a) and (c)] and GCN [(b) and (d)].

capability of graph-based models to maximize leveraging the available structural features and learn correlations without requiring more than information on molecular structure in the form of a graph for accurate property prediction. Therefore, the graph-based models' mechanism to update embeddings makes them suitable for predicting molecular properties from molecular structure. This results in better performance metrics for graph-based models than linear regression.

The graph-based models presented here are accurate, robust, and invariant in dataset size and origin. The models are highly versatile and are demonstrated to predict a wide range of properties with high accuracy rather than being tuned for specific properties, thereby more effectively evaluating molecules for various applications. The model is versatile, wherein it predicts ground state properties such as orbital energies and excited state properties, including optical absorption and emission, with high accuracy based on molecular structure. In this way, the model can analyze the performance of a given molecule considering these various aspects, thereby allowing the design of optimal molecules for a given application. Further, the models are versatile across classes of properties. We show hierarchical pooling implemented in the convolution layer to enhance the prediction of excited state properties. This enhancement in prediction is because the hierarchical pooling method allows only the nodes with more significant contributions selected by a scoring system to undergo message passing in

subsequent cycles. The additional weighing of the contribution of nodes leads to updated embeddings that better correlate with the excited state properties. This implies that a subset of the molecular composition has a more significant influence on the excited state properties of the molecules. Implementing hierarchical pooling adds to the model a mechanism to identify the subset by effectively weighing its contribution as larger while generating $\vec{G}$, leading to enhanced learning and prediction of excited state properties. In contrast, there is greater compression of information in the algorithm using the global pooling method. The updated values of nodes are converted to molecular embeddings directly by the global pooling method; hence, correlations with excited state properties are not learned as effectively. With this, the same model is proven to predict ground and excited state properties accurately.

Previously, general-purpose models have been developed but have usually not included the prediction of orbital energies along with optical absorption or emission wavelengths.^{45–47} Moreover, the graph-based models presented here achieve a higher level of accuracy. For example, MAE was reported by Burns *et al.*⁴⁶ for a general-purpose message-passing-based model, with 0.306 eV for HOMO and 0.258 eV for LUMO. The GAT model developed in this work has achieved an MAE of 0.16 eV for HOMO and 0.11 eV for LUMO. Similarly, the lowest MAE reported for IP and VEE in ref. 45 are respectively 0.173 and 0.249, and the respective r^2 are 0.87 and 0.76. The GAT model has achieved

MAE of 0.18 and 0.13, respectively, for IP and VEE, while the respective r^2 are 0.95 and 0.87. We attribute this high accuracy to graph-based models' capacity to extract crucial information from molecular structures to correlate with molecular properties.

5 Conclusion

Model architectures are proposed based on graph-based networks optimized for predicting diverse molecular properties, including both ground state and excited state properties, based on minimal molecular features extracted only from SMILES strings of molecules as input. The algorithms based on GAT and GCN mechanisms are trained and tested for ground state and excited state molecular properties against experimentally measured or computed values from two independent datasets. Model architectures were optimized with fixed input features and convolution networks of fixed dimensions. The models were trained with respect to different properties to obtain optimized hyperparameters for each. We demonstrate GAT based model to be consistently accurate compared to other methods. Graph-based methods are robust and have high prediction accuracy, even for small-sized data sets. While prediction by RF loses accuracy with smaller data set sizes, graph-based models are efficient in all cases, attributed to the efficient processing of embeddings in the convolution layers leading to better correlations with target properties. The marginal differences between the accuracy of GAT and GCN indicate that the attention mechanisms involved in GAT offer only a slight advantage in predicting molecular properties from structure. However, the advantage is consistently observed across all ground and excited state properties. A hierarchical pooling method was implemented with the convolution layers to impart versatility to the model, allowing nodes to be weighed while updating embeddings. We show that this strategy enhanced the prediction accuracy of excited state properties by effectively selecting the most influential nodes. Thus, versatile graph-based models presented here are efficient tools for molecular screening and design for applications relying on multiple molecular properties.

Data availability

Datasets employed in the study can be obtained at Virtual Excited State Reference for the Discovery of Electronic Materials Database (VERDE Materials DB) accessible at URL "<https://www.verdematerialsdb.com/>", and Nature Scientific database accessible at DOI <https://doi.org/10.1038/s41597-020-00634-8>. The code will be made available upon reasonable request.

Conflicts of interest

The authors have no conflicts to declare.

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